Unit I :: Distribution of Velocities and Molecular Collisions

☙ PHY0500304: Heat & Thermodynamics ❧

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Syllabus

Unit I: Distribution of Velocities and Molecular Collisions

Maxwell-Boltzmann Law of Distribution of Velocities in an Ideal Gas and its Experimental Verification. Mean, RMS and Most Probable Speeds. Degrees of Freedom. Law of Equipartition of Energy (No proof required). Mean Free Path. Collision Probability. Transport Phenomenon in Ideal Gases: (1) Viscosity, and (2) Thermal Conductivity. Brownian Motion (qualitative idea only).

1 Maxwell-Boltzmann distribution law

The Maxwell-Boltzmann distribution is a result of the kinetic theory of gases, which provides a simplified explanation of many fundamental gaseous properties, including pressure and diffusion. This distribution applies fundamentally to particle velocities in three dimensions, but turns out to depend only on the speed (the magnitude of the velocity) of the particles. A particle speed probability distribution indicates which speeds are more likely: a randomly chosen particle will have a speed selected randomly from the distribution, and is more likely to be within one range of speeds than another.

The kinetic theory of gases applies to the classical ideal gas, which is an idealization of real gases. In real gases, there are various effects (e.g., van der Waals interactions, vortical flow, relativistic speed limits, and quantum exchange interactions) that can make their speed distribution different from the Maxwell–Boltzmann form. However, rarefied gases at ordinary temperatures behave very nearly like an ideal gas and the Maxwell speed distribution is an excellent approximation for such gases. This is also true for ideal plasmas, which are ionized gases of sufficiently low density.

1.1 Postulates of kinetic theory of gases

1. Every mole of a gas contains about 6.023×10^{-23} molecules.
2. The gas molecules may be regarded as point masses.
3. The gas molecules are in a state of continuous random motion.
4. In the absence of an external force field, the molecules are distributed uniformly in the container, i.e. an ideal gas behaves as an isotropic medium.
5. The molecules of a gas experience force only during collisions.
6. The molecules of a gas behave as rigid, perfectly elastic hard spheres.
7. The duration of collisions is negligible compared to the time interval between collisions.
8. There is a spread of molecular speeds from zero to infinity.
9. The gravitational potential energy has negligible effect on the motion of the gas molecules.

1.2 Fundamental assumptions

In the derivation of the Maxwell-Boltzmann distribution, we make the following basic assumptions:

1. In the equilibrium state, the molecules have complete randomness of direction and velocity.
2. There is no mass motion or convection current in the body of the gas.

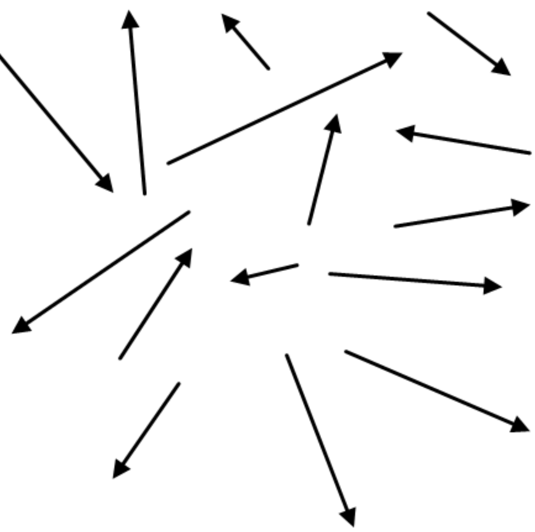
3. The probability of a molecule having an x-velocity component at any point of time is independent of its y- and z-velocity components, and so on.

4. The probability that a molecule selected at random has velocity component in a given range is a function only of the magnitude of the velocity component and the width of the interval.

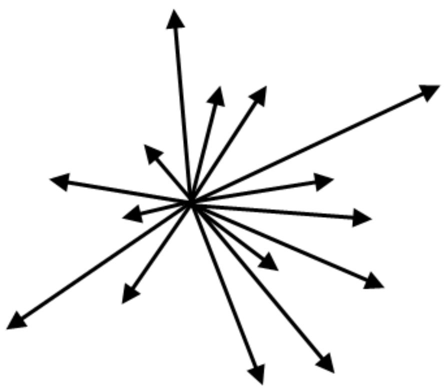
5. The gas molecules have no rotational or vibrational energies.

1.3 Velocity space

Consider a gas of N molecules enclosed in a vessel, of arbitrary shape, and moving randomly. The velocity vectors are all randomly oriented in all possible directions.



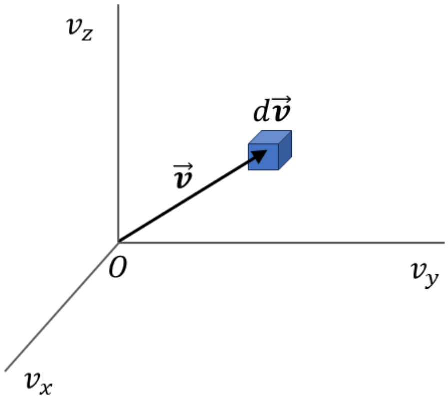
Now we transport each vector parallel to itself to a common origin.



If we now fix the velocity axes v_x, v_y and v_z with respect to a common origin O (that corresponds to particles at rest), every molecular velocity can be represented in this space, which is known as the *velocity space*.

1.4 Derivation of Maxwell-Boltzmann distribution law

Let us now consider an infinitessimal volume element $\mathrm{d}\vec{\boldsymbol{v}}$ in this velocity space. The number of velocity vectors that terminate inside $\mathrm{d}\vec{\boldsymbol{v}}$ gives us the average number of molecules whose velocities lie between $\vec{\boldsymbol{v}}$ and $\vec{\boldsymbol{v}}+\mathrm{d}\vec{\boldsymbol{v}}$.



We have, for any velocity $\vec{\boldsymbol{v}}$, the corresponding speed to be

$$v^2 = v_x^2 + v_y^2 + v_z^2 \tag{1}$$

Let $\mathrm{d}N_{v_x}$ be the number of molecules having velocity components in the range v_x and $v_x+\mathrm{d}v_x$. Then from assumption #4, we have

$$\frac{\mathrm{d}N_{v_x}}{N} = f(v_x)\,\mathrm{d}v_x$$
$$\text{or, } \mathrm{d}N_{v_x} = N f(v_x)\,\mathrm{d}v_x \tag{2}$$

In equation (2), $f(v_x)$ is an unknown function that we have to determine.

Also, the probability of finding a molecule with x-velocity component in the range v_x and v_x+$\mathrm{d}v_x$ is

$$p_x = \frac{\mathrm{d}N_{v_x}}{N} = f(v_x)\,\mathrm{d}v_x \tag{3}$$

Likewise, from assumption #3, we have

$$p_y = f(v_y)\,\mathrm{d}v_y \tag{4}$$
$$p_z = f(v_y)\,\mathrm{d}v_z \tag{5}$$

Then from the law of compound probabilities, the probability of finding a molecule having velocity components simultaneously in the ranges $[v_x, v_x+\mathrm{d}v_x]$, $[v_y, v_y+\mathrm{d}v_y]$ and $[v_z, v_z+\mathrm{d}v_z]$ is

$$\frac{\mathrm{d}^3 N_{v_x v_y v_z}}{N} = p_x p_y p_z = f(v_x) f(v_y) f(v_z)\,\mathrm{d}v_x\,\mathrm{d}v_y\,\mathrm{d}v_z \tag{6}$$

Hence, the number density of velocity points is

$$\rho = \frac{\mathrm{d}^3 N_{v_x v_y v_z}}{\mathrm{d}v_x\,\mathrm{d}v_y\,\mathrm{d}v_z} = N f(v_x) f(v_y) f(v_z) \tag{7}$$

It follows from assumption #1 that the velocity space has been assumed to be isotropic, and so this implies that ρ is independent of the direction of $\vec{v}$ and only depends ton v^2, i.e

$$\begin{aligned} \rho &\propto J(v^2) \\ \implies f(v_x)f(v_y)f(v_z) &= J(v^2) \end{aligned} \tag{8}$$

where J is some other function.

For a fixed value of v^2, $J(v^2)$ is constant and so

$$\begin{aligned} \mathrm{d}\big[J(v^2)\big] &= 0 \\ \implies \mathrm{d}\big[f(v_x)f(v_y)f(v_z)\big] &= 0 \\ \implies \frac{\partial f(v_x)}{\partial v_x}\mathrm{d}v_x\, f(v_y)f(v_z) + f(v_x)\frac{\partial f(v_y)}{\partial v_y}\mathrm{d}v_y\, f(v_z) + f(v_x)f(v_y)\frac{\partial f(v_z)}{\partial v_z}\mathrm{d}v_z &= 0 \end{aligned} \tag{9}$$

Dividing equation (9) by $f(v_x)f(v_y)f(v_z)$, we obtain

$$\frac{1}{f(v_x)}\frac{\partial f(v_x)}{\partial v_x}\mathrm{d}v_x + \frac{1}{f(v_y)}\frac{\partial f(v_y)}{\partial v_y}\mathrm{d}v_y + \frac{1}{f(v_z)}\frac{\partial f(v_z)}{\partial v_z}\mathrm{d}v_z = 0 \tag{10}$$

Also, for a fixed value of v^2, we have

$$\begin{aligned} \mathrm{d}\big[v^2\big] &= 0 \\ \implies \mathrm{d}\big[v_x^2 + v_y^2 + v_z^2\big] &= 0 \\ \implies v_x\,\mathrm{d}v_x + v_y\,\mathrm{d}v_y + v_z\,\mathrm{d}v_z &= 0 \end{aligned} \tag{11}$$

Suppose that B is some unknown constant. Then multiplying equation (11) with $2B$ and adding to equation (10) gives

$$\left[\frac{1}{f(v_x)}\frac{\partial f(v_x)}{\partial v_x} + 2Bv_x\right]\mathrm{d}v_x + \left[\frac{1}{f(v_y)}\frac{\partial f(v_y)}{\partial v_y} + 2Bv_y\right]\mathrm{d}v_y + \left[\frac{1}{f(v_z)}\frac{\partial f(v_z)}{\partial v_z} + 2Bv_z\right]\mathrm{d}v_z = 0 \tag{12}$$

Even though v is held constant, v_x, v_y, v_z may individually change and yet satisfy equation (1). In equation (12), the first term within brackets depends only on v_x, the second term on v_y and the third term on v_z.

Because $\mathrm{d}v_x, \mathrm{d}v_y, \mathrm{d}v_z$ are arbitrary, the only way for equation (12) to hold good is if all three terms vanish identically.

Therefore, for the v_x term, we would have

$$\begin{aligned} \frac{1}{f(v_x)}\frac{\partial f(v_x)}{\partial v_x} + 2Bv_x &= 0 \\ \implies \frac{\partial f(v_x)}{\partial v_x} &= -2Bv_x f(v_x) \\ \implies \frac{\partial f(v_x)}{f(v_x)} &= -2Bv_x \partial v_x \\ \implies \int \frac{\mathrm{d}f}{f} &= -2B\int v_x\,\mathrm{d}v_x \\ \implies \ln f(v_x) &= -2B\frac{v_x^2}{2} + \ln A \end{aligned}$$

where $\ln A$ is the constant of integration. From the above we obtain

$$\begin{aligned}
\ln f(v_x) - \ln A &= -Bv_x^2 \\
\implies \ln \frac{f(v_x)}{A} &= -Bv_x^2 \\
\implies \frac{f(v_x)}{A} &= e^{-Bv_x^2} \\
\implies f(v_x) &= Ae^{-Bv_x^2} \qquad (13)
\end{aligned}$$

Likewise, from the v_y and v_z terms, we obtain

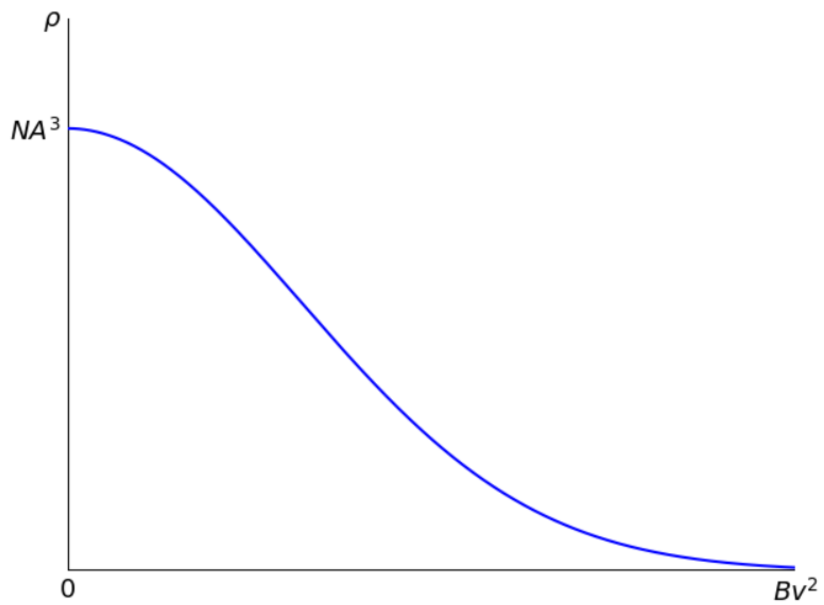
$$f(v_y) = Ae^{-Bv_y^2} \qquad (14)$$

$$f(v_z) = Ae^{-Bv_z^2} \qquad (15)$$

Using equations (13), (14) and (15) in equation (7) gives us

$$\begin{aligned}
\rho &= N\left(Ae^{-Bv_x^2}\right)\left(Ae^{-Bv_y^2}\right)\left(Ae^{-Bv_z^2}\right) \\
&= NA^3 e^{-B(v_x^2+v_y^2+v_z^2)} \\
\implies & \boxed{\rho(v) = NA^3 e^{-Bv^2}} \qquad (16)
\end{aligned}$$

Equation (16) is known as *Maxwell's velocity density distribution function*, where ρ varies with molecular speeds v as shown in the plot below:



1.4.1 Molecular distribution of speeds

1.4.2 Determination of the constants

2 Molecular speeds

2.1 Average or mean speed

2.2 Root mean squared (RMS) speed

2.3 Most probable speed

3 Energy distribution of a Maxwellian gas

Insert section content.

4 Molecular energies

4.1 Average energy

4.2 Most probable energy

5 Molecular collisions

5.1 Mean free path